

# Proof ledger

Repository source: proof/PROOF\_LEDGER.md

## Status vocabulary

- **ESTABLISHED IN MANUSCRIPT:** a complete derivation is present in the paper, subject to ordinary mathematical review.
- **EXACT-CERTIFIED:** the finite claim is checked with integer and rational arithmetic by the standalone verifier.
- **NUMERICAL AUDIT:** a separate floating-point implementation agrees.
- **RECONSTRUCTION-CHECKED:** an expected derived artifact is rebuilt byte for byte.
- **MUTATION-TESTED:** a targeted corrupted object is rejected.
- **SOURCE-CHECKED:** bibliographic or historical wording has been compared with an identified source.
- **HOSTED-REPLAYED:** the named three-job hosted workflow passed on the stated commit.
- **FRESH-CLONE-REPLAYED:** a new clone of the remote passed the stated replay and ended clean.
- **RELEASE-PENDING:** technically prepared but awaiting a final evidence run, immutable tag, release assets, visibility change, or explicit approval.
- **EXTERNAL REVIEW NEEDED:** no independent subject-matter review has been recorded.

## Mathematical claims

| ID | Claim                                                                | Evidence                                 | Status                                        |
|----|----------------------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| M1 | Constant-width support identity and diameter-one normalization       | Paper Section 2; classical references    | ESTABLISHED IN MANUSCRIPT;<br>SOURCE-CHECKED  |
| M2 | Blaschke volume-surface identity                                     | Paper Section 2; classical references    | ESTABLISHED IN MANUSCRIPT;<br>SOURCE-CHECKED  |
| M3 | Minimum circumsphere contact balance and Jung range                  | Paper Section 2                          | ESTABLISHED IN MANUSCRIPT                     |
| M4 | Opposite-point identity and radial support inequality                | Paper Lemma 2.1                          | ESTABLISHED IN MANUSCRIPT                     |
| L1 | Transport inequality and tangent-gap master inequality               | Paper Section 3                          | ESTABLISHED IN MANUSCRIPT                     |
| L2 | Rational propagation implications                                    | Paper Section 3; exact lower-cell checks | ESTABLISHED IN MANUSCRIPT;<br>EXACT-CERTIFIED |
| L3 | Four refined cells replace 03A1 with positive margins                | Refined NPZ cells and JSON               | EXACT-CERTIFIED;<br>NUMERICAL AUDIT           |
| L4 | Complete lower partition is gap free and has unique bottleneck 03B02 | Final JSON and exact verifier            | EXACT-CERTIFIED;<br>RECONSTRUCTION-CHECKED    |
| N1 | Near-Jung four-contact deficiency budget                             | Paper Section 4                          | ESTABLISHED IN MANUSCRIPT                     |
| N2 | Affine contact coordinates and Gram formulas                         | Paper Section 4                          | ESTABLISHED IN MANUSCRIPT                     |

| ID | Claim                                                                | Evidence                                                                            | Status                                                               |
|----|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| N3 | Determinant transfer<br>$\det H \geq 1 - S/2 - S^2/2$                | Paper Proposition 4.1                                                               | ESTABLISHED IN<br>MANUSCRIPT; exact<br>endpoint evaluation           |
| N4 | Dominant-edge<br>coordinate-width diagonal<br>bounds                 | Paper Section 5                                                                     | ESTABLISHED IN<br>MANUSCRIPT; exact<br>endpoint evaluation           |
| N5 | Adjusted 98,306-ray fan is<br>feasible with positive exact<br>margin | Fan NPZ, orbit TSV,<br>exact verifier                                               | EXACT-CERTIFIED;<br>NUMERICAL AUDIT;<br>MUTATION-TESTED              |
| N6 | Forced-core volume lower<br>bound                                    | Exact fan volume and<br>determinant transfer                                        | EXACT-CERTIFIED;<br>NUMERICAL AUDIT                                  |
| N7 | Three axis-specific<br>antipodal cap-pair bounds                     | Hull TSV and cap<br>certificate fields                                              | EXACT-CERTIFIED;<br>NUMERICAL AUDIT;<br>MUTATION-TESTED              |
| N8 | Six cap interiors are<br>pairwise disjoint                           | Paper and exact<br>disjointness fields                                              | ESTABLISHED IN<br>MANUSCRIPT;<br>EXACT-CERTIFIED;<br>MUTATION-TESTED |
| N9 | Near-Jung lower bound<br>exceeds theorem upper<br>enclosure          | Final JSON assembly                                                                 | EXACT-CERTIFIED;<br>NUMERICAL AUDIT                                  |
| T1 | Universal coefficient<br>equals the exact theorem<br>coefficient     | 03B02 plus strict near-Jung<br>handoff; exact value in the<br>certificate and paper | EXACT-CERTIFIED;<br>RECONSTRUCTION-<br>CHECKED                       |
| T2 | Scaling gives the theorem<br>for arbitrary width $d > 0$             | Homogeneity                                                                         | ESTABLISHED IN<br>MANUSCRIPT                                         |

## Verification and provenance claims

| ID | Claim                                                              | Evidence                                      | Status                     |
|----|--------------------------------------------------------------------|-----------------------------------------------|----------------------------|
| V1 | Standalone verifier has no<br>private repository<br>dependency     | Path regression test and<br>source inspection | EXACT-CERTIFIED            |
| V2 | Final JSON is rebuilt byte<br>identically                          | Exact verifier and unit test                  | RECONSTRUCTION-<br>CHECKED |
| V3 | Exact TSV and CSV<br>renderings are rebuilt byte<br>identically    | Exact verifier                                | RECONSTRUCTION-<br>CHECKED |
| V4 | Binary64 implementation<br>agrees on principal<br>quantities       | Binary64 transcript                           | NUMERICAL AUDIT            |
| V5 | Base-R implementation<br>agrees on principal<br>quantities         | Named-author and hosted<br>R transcripts      | NUMERICAL AUDIT            |
| V6 | Seven named proof-object<br>mutations are rejected                 | Mutation transcript                           | MUTATION-TESTED            |
| V7 | Paper-facing constants<br>and lower table match the<br>JSON        | Paper reconciliation script                   | RECONSTRUCTION-<br>CHECKED |
| V8 | Exact-constants appendix<br>matches the JSON                       | Generator <code>--check</code>                | RECONSTRUCTION-<br>CHECKED |
| V9 | Required release<br>documents exist and<br>preserve claim language | Release-document checker                      | RECONSTRUCTION-<br>CHECKED |

| ID  | Claim                                                                              | Evidence                                                                                            | Status                                       |
|-----|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------|
| V10 | Markdown/PDF pairs are complete                                                    | Pair checker                                                                                        | RECONSTRUCTION-CHECKED                       |
| V11 | PDFs are readable, have embedded fonts, and expose text                            | Hosted Poppler preflight on run 35468019663                                                         | HOSTED-REPLAYED                              |
| V12 | Exact, Python-audit, test, mutation, metadata, and paper paths pass on Ubuntu      | Workflow run 35468019663, commit 892d4bc6fad07950e78da66e45b95063a6415af6                           | HOSTED-REPLAYED                              |
| V13 | Independent base-R audit passes on hosted R 4.6.1                                  | Workflow run 35468019663                                                                            | HOSTED-REPLAYED; NUMERICAL AUDIT             |
| V14 | Remote fresh clone reproduces all language paths and builds                        | Named-author transcript for commit 892d4bc6...                                                      | FRESH-CLONE-REPLAYED                         |
| V15 | Fresh-clone working tree remains clean and bundle has complete five-commit history | Fresh transcript and verified transfer bundle                                                       | FRESH-CLONE-REPLAYED; RECONSTRUCTION-CHECKED |
| V16 | Hosted/fresh-clone evidence fields and claim boundaries are indexed consistently   | <code>certificate/evidence/rc1_remote_audit.json</code> ; <code>ci/verify_remote_evidence.py</code> | RECONSTRUCTION-CHECKED                       |

The hosted evidence above refers to <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>.

## Publication and review claims

| ID  | Claim                                                                  | Status                          |
|-----|------------------------------------------------------------------------|---------------------------------|
| P1  | Lean ten-page Paper v1 manuscript is present                           | Completed draft                 |
| P2  | Complete release-document set is present in Markdown and PDF           | Completed; 22 paired documents  |
| P3  | Release-candidate author list and no-affiliation statement are fixed   | Completed for v1.0.0-rc1        |
| P4  | Split license and release-candidate citation metadata are fixed        | Completed for v1.0.0-rc1        |
| P5  | Private staging remote exists and the audited candidate was pushed     | Completed at commit 892d4bc6... |
| P6  | Hosted Python, R, and document jobs are green on the audited candidate | Completed on run 35468019663    |
| P7  | Fresh staging-remote clone replay is recorded                          | Completed; final status clean   |
| P8  | Evidence-bearing commit receives its own hosted run                    | RELEASE-PENDING                 |
| P9  | Immutable final tag and release assets exist                           | RELEASE-PENDING                 |
| P10 | Repository visibility is public and GitHub Release is published        | RELEASE-PENDING                 |
| P11 | External subject-matter reviewer has checked the theorem               | EXTERNAL REVIEW NEEDED          |

P12

Peer review or journal acceptance  
exists

Not claimed

## Claim boundary

The package does not establish equality, rigidity, sharpness, a minimizing body, or Meissner extremality. It should not be described as independently verified or formally verified while P11 remains open. The named-author fresh clone and hosted CI are reproducibility evidence, not external mathematical review.

## Current bottleneck

The mathematical, finite-certificate, authorship, citation, licensing, hosted- candidate, and fresh-clone gates are complete for audited commit `892d4bc6fad07950e78da66e45b95063a6415af6`. The remaining release bottlenecks are:

1. green hosted jobs on this evidence-bearing commit;
2. final named-author copy edit and page approval;
3. final release date and transition to `v1.0.0`;
4. immutable release assets and independent asset-hash verification;
5. annotated `v1.0.0` tag, green tag workflow, and fresh tagged-checkout replay;
6. explicit approval of the final README, paper, release notes, tag, visibility change, release, and announcement;  
and
7. external mathematical review after release.